COMPACT OBJECTS OF THE DERIVED QUASI-COHERENT CATEGORY ARE EXACTLY THE PERFECT ONES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof establishes compactness of perfect objects through the duality formula with direct-sum commutation, then reduces the converse to affine etale opens via Rj_* direct-sum commutation and finally to the scheme case via the comparison lemmas

Editorial note. The label proposition-compact-is-perfect names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- the duality formula Hom_{D(O_X)}(K, M) = H^0(X, K^vee tensor^L M) for K perfect with dual K^vee (Cohomology on Sites, Lemma sites-cohomology-lemma-dual-perfect-complex)
- commutation of tensoring with a perfect complex with direct sums, and commutation of H^0(X, -) with direct sums on D_QCoh(O_X) (lemma-quasi-coherence-pushforward-direct-sums of the same chapter)
- perfectness of an object of D_QCoh checked etale locally (Cohomology on Sites, Lemma sites-cohomology-lemma-perfect-independent-representative)
- Rj_* commutes with direct sums for quasi-compact separated morphisms j (lemma-quasi-coherence-pushforward-direct-sums of the same chapter) and the adjunction between restriction and Rj_*
- comparison of D_QCoh on the small etale site with the scheme case (lemma-derived-quasi-coherent-small-etale-site and lemma-descend-perfect of the same chapter) and the scheme proposition perfect-proposition-compact-is-perfect of Derived Categories of Schemes

Source reference. Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at proof line(s) 4133; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08HG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3288–3288. Prerequisite pointer invoked at proof line(s) 4134; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 08DZ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 863–863. Prerequisite pointer invoked at proof line(s) 4115; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 09M1: sources/stacks-chow/tags.txt, source lines 12164–12164. Prerequisite pointer invoked at proof line(s) 4137; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08JJ: sources/stacks-chow/tags.txt, source lines 10852–10852. Prerequisite pointer invoked at proof line(s) 4106; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08G6: sources/stacks-chow/tags.txt, source lines 10734–10734. Prerequisite pointer invoked at proof line(s) 4123; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. An object of $D_{QCoh}(\mathcal{O}_X)$ is compact if and only if it is perfect.*

Proof. If K is a perfect object of $D(\mathcal{O}_X)$ with dual $K^\vee$ (Cohomology on Sites, Lemma 08JJ) we have

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K, M) = H^0(X, K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} M)$$

functorially in M. Since $K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} -$ commutes with direct sums and since $H^0(X, -)$ commutes with direct sums on $D_{QCoh}(\mathcal{O}_X)$ by Lemma 08DZ we conclude that K is compact in $D_{QCoh}(\mathcal{O}_X)$.

Conversely, let K be a compact object of $D_{QCoh}(\mathcal{O}_X)$. To show that K is perfect, it suffices to show that $K|_U$ is perfect for every affine scheme U étale over X, see Cohomology on Sites, Lemma 08G6. Observe that $j : U \to X$ is a quasi-compact and separated morphism. Hence $Rj_* : D_{QCoh}(\mathcal{O}_U) \to D_{QCoh}(\mathcal{O}_X)$ commutes with direct sums, see Lemma 08DZ. Thus the adjointness of restriction to U and Rj_* implies that $K|_U$ is a perfect object of $D_{QCoh}(\mathcal{O}_U)$. Hence we reduce to the case that X is affine, in particular a quasi-compact and quasi-separated scheme. Via Lemma 071Q and 08HG we reduce to the case of schemes, i.e., to Derived Categories of Schemes, Proposition 09M1. □